

Table A
Bias Correction Factors
For Estimating Standard Deviations

Subgroup Size	d_2	c_2	c_4	d_3	Subgroup Size	d_2	c_2	c_4	d_3
2	1.128	0.5642	.7979	0.8525	21	3.778	0.9638	.9876	0.7272
3	1.693	0.7236	.8862	0.8884	22	3.819	0.9655	.9882	0.7199
4	2.059	0.7979	.9213	0.8798	23	3.858	0.9670	.9887	0.7159
5	2.326	0.8407	.9400	0.8641	24	3.895	0.9684	.9892	0.7121
6	2.534	0.8686	.9515	0.8480	25	3.931	0.9695	.9896	0.7084
7	2.704	0.8882	.9594	0.8332	30	4.086	0.9748	.9915	0.6927
8	2.847	0.9027	.9650	0.8198	35	4.213	0.9784	.9927	0.6799
9	2.970	0.9139	.9693	0.8078	40	4.322	0.9811	.9936	0.6692
10	3.078	0.9227	.9727	0.7971	45	4.415	0.9832	.9943	0.6601
11	3.173	0.9300	.9754	0.7873	50	4.498	0.9849	.9949	0.6521
12	3.258	0.9359	.9776	0.7785	60	4.639	0.9874	.9957	0.6389
13	3.336	0.9410	.9794	0.7704	70	4.755	0.9892	.9963	0.6283
14	3.407	0.9453	.9810	0.7630	80	4.854	0.9906	.9968	0.6194
15	3.472	0.9490	.9823	0.7562	90	4.939	0.9916	.9972	0.6118
16	3.532	0.9523	.9835	0.7499	100	5.015	0.9925	.9975	0.6052
17	3.588	0.9551	.9845	0.7441					
18	3.640	0.9576	.9854	0.7386					
19	3.689	0.9599	.9862	0.7335					
20	3.735	0.9619	.9869	0.7287					

"Unbiased" estimates of various Standard Deviations will be obtained from any of the following:

$$\text{An Unbiased Est. of SD}(X) = \frac{s_n}{c_2} \text{ or } \frac{\bar{s}_n}{c_2}$$

$$\text{An Unbiased Est. of SD}(X) = \frac{s}{c_4} \text{ or } \frac{\bar{s}}{c_4}$$

$$\text{An Unbiased Est. of SD}(X) = \frac{R}{d_2} \text{ or } \frac{\bar{R}}{d_2}$$

$$\text{An Unbiased Est. of SD}(R) = \frac{d_3 \bar{R}}{d_2}$$

TABLE 7.2. GUIDE FOR CONSTRUCTING CONTROL CHARTS BASED ON VARIABLES

Quantity controlled	Central line	Lower control limit (LCL)	Upper control limit (UCL)	Sample size	Remarks
Mean $\bar{x}$	$\bar{\bar{x}}$	$\bar{\bar{x}} - A_2\bar{R}$	$\bar{\bar{x}} + A_2\bar{R}$	Small, preferably 10 or fewer	Uses range R . No standard μ and σ given.
Sum S	$\bar{S}$	$\bar{S} - A_0\bar{R}$	$\bar{S} + A_0\bar{R}$	Small, preferably 10 or fewer	No standard μ and σ given.
Mean $\bar{x}$	$\bar{\bar{x}}$	$\bar{\bar{x}} - \bar{A}_1\sqrt{\frac{n-1}{n}}$	$\bar{\bar{x}} + \bar{A}_1\sqrt{\frac{n-1}{n}}$	25 or fewer, constant size	No standard μ and σ given.
Mean $\bar{x}$	$\bar{\bar{x}}$	$\bar{\bar{x}} - 3\bar{s}/\sqrt{n}$	$\bar{\bar{x}} + 3\bar{s}/\sqrt{n}$	Sample sizes ≥ 25 , may vary slightly	No standard μ and σ given.
Mean $\bar{x}$	μ	$\mu - A\sigma$	$\mu + A\sigma$	n	μ and σ specified (standard values are to be maintained).
Range R	$\bar{R}$	$D_3\bar{R}$	$D_4\bar{R}$	Small, preferably 10 or fewer	No standard σ given.
Range R	$d_2\sigma$	$D_3\sigma$	$D_4\sigma$	Small, preferably 10 or fewer	σ specified (standard value to be maintained).
Sample standard deviation s	$\bar{s}$	$B_3\bar{s}$	$B_4\bar{s}$	25 or fewer, constant size	No standard σ given.
Sample standard deviation s	$\bar{s}$	$\bar{s} - 3\bar{s}/\sqrt{2n}$	$\bar{s} + 3\bar{s}/\sqrt{2n}$	Sample sizes ≥ 25 , may vary slightly	No standard σ given.
Sample standard deviation s	$\sigma c_2\sqrt{\frac{n}{n-1}}$	$\sigma B_1\sqrt{\frac{n}{n-1}}$	$\sigma B_2\sqrt{\frac{n}{n-1}}$	n	σ specified (standard value to be maintained).

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TABLE 18. FACTORS FOR COMPUTING CONTROL-CHART LINES *

Sample size n	Chart for averages				Chart for standard deviations								Chart for ranges			
	Factors for control limits				Factors for central line		Factors for control limits				Factors for central line		Factors for control limits			
					c_2	$1/c_2$	B_1	B_2	B_3	B_4	d_2	$1/d_2$	D_1	D_2	D_3	D_4
2	2.121	3.760	3.760	1.880	0.5642	1.7725	0	1.843	0	3.267	1.128	0.8865	0	3.686	0	3.267
3	1.732	3.070	2.394	1.023	0.7236	1.3820	0	1.858	0	2.568	1.693	0.5907	0	4.358	0	2.575
4	1.500	2.914	1.880	0.729	0.7979	1.2533	0	1.808	0	2.266	2.059	0.4857	0	4.698	0	2.282
5	1.342	2.884	1.596	0.577	0.8407	1.1894	0	1.756	0	2.089	2.326	0.4299	0	4.918	0	2.115
6	1.225	2.899	1.410	0.483	0.8686	1.1512	0.026	1.711	0.030	1.970	2.534	0.3946	0	5.078	0	2.004
7	1.134	2.935	1.277	0.419	0.8882	1.1259	0.105	1.672	0.118	1.882	2.704	0.3698	0.205	5.203	0.076	1.924
8	1.061	2.980	1.175	0.373	0.9027	1.1078	0.167	1.638	0.185	1.815	2.847	0.3512	0.387	5.307	0.136	1.864
9	1.000	3.030	1.094	0.337	0.9139	1.0942	0.219	1.609	0.239	1.761	2.970	0.3367	0.546	5.394	0.184	1.816
10	0.949	3.085	1.028	0.308	0.9227	1.0837	0.262	1.584	0.284	1.716	3.078	0.3249	0.687	5.469	0.223	1.777
11	0.905	3.136	0.973	0.285	0.9300	1.0753	0.299	1.561	0.321	1.679	3.173	0.3152	0.812	5.534	0.256	1.744
12	0.866	3.189	0.925	0.266	0.9359	1.0684	0.331	1.541	0.354	1.646	3.258	0.3069	0.924	5.592	0.284	1.716
13	0.832	3.242	0.884	0.249	0.9410	1.0627	0.359	1.523	0.382	1.618	3.336	0.2998	1.026	5.646	0.308	1.692
14	0.802	3.295	0.848	0.235	0.9453	1.0579	0.384	1.507	0.406	1.594	3.407	0.2935	1.121	5.693	0.329	1.671
15	0.775	3.347	0.816	0.223	0.9490	1.0537	0.406	1.492	0.428	1.572	3.472	0.2880	1.207	5.737	0.348	1.652
16	0.750	3.398	0.788	0.212	0.9523	1.0501	0.427	1.478	0.448	1.552	3.532	0.2831	1.285	5.779	0.364	1.636
17	0.728	3.448	0.762	0.203	0.9551	1.0470	0.445	1.465	0.466	1.534	3.588	0.2787	1.359	5.817	0.379	1.621
18	0.707	3.497	0.738	0.194	0.9576	1.0442	0.461	1.454	0.482	1.518	3.640	0.2747	1.426	5.854	0.392	1.608
19	0.688	3.545	0.717	0.187	0.9599	1.0418	0.477	1.443	0.497	1.503	3.689	0.2711	1.490	5.888	0.404	1.596
20	0.671	3.592	0.697	0.180	0.9619	1.0396	0.491	1.433	0.510	1.490	3.735	0.2677	1.548	5.922	0.414	1.586
21	0.655	3.639	0.679	0.173	0.9638	1.0376	0.504	1.424	0.523	1.477	3.778	0.2647	1.606	5.950	0.425	1.575
22	0.640	3.684	0.662	0.167	0.9655	1.0358	0.516	1.415	0.534	1.466	3.819	0.2618	1.659	5.979	0.434	1.566
23	0.626	3.729	0.647	0.162	0.9670	1.0342	0.527	1.407	0.545	1.455	3.858	0.2592	1.710	6.006	0.443	1.557
24	0.612	3.773	0.632	0.157	0.9684	1.0327	0.538	1.399	0.555	1.445	3.895	0.2567	1.759	6.031	0.452	1.548
25	0.600	3.816	0.619	0.153	0.9696	1.0313	0.548	1.392	0.565	1.435	3.931	0.2544	1.804	6.058	0.459	1.541
> 25	$\frac{3}{\sqrt{n}}$		$\frac{3}{\sqrt{n}}$		1.0000	1.0000	••	•••	••	•••						

* Use explained in Chapter 7. The relation $A_0 = 3\sqrt{n}/d_2$ holds. Adapted, with the permission of the American Society for Testing Materials, from ASTM Manual on Quality Control of Materials, Philadelphia, January 1951, p. 115.

$$•• 1 - \frac{3}{\sqrt{2n}}$$

$$••• 1 + \frac{3}{\sqrt{2n}}$$

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Source : E.L. Crow, F.A. Davis, M.W. Maxfield, "Statistics Manual", Dover Publications, New York, 1960.